

Interim Removable Partial Denture

Learning objectives

By the end of this lecture, you will be able to:

- Define the interim RPD and list its key indications.
 - Describe its main components and design features.
 - Explain the function and types of clasps used.
 - Identify the potential drawbacks of interim RPDs.
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What is an Interim RPD?

An interim removable partial denture (RPD) is sometimes fabricated before a definitive removable partial denture is made. The patient may wear the interim RPD for a short period or a longer period, depending on the situation.

Other terms to describe the interim removable partial dentures are: flipper, temporary, transitional, treatment, provisional or Interim.

When Do We Use an Interim RPD? - The Indications

1. **Young age** (less than 17 years)- arches are still growing, and a permanent prosthesis would not be suitable for long. Large pulps may preclude the use of fixed partial dentures until a later time.
2. **Transitional Denture:** This allows the patient to gradually adapt to having fewer teeth and a prosthesis, making the final transition to a complete denture less traumatic.
 - (a) Psychological Preparation: Transitional denture for patients not ready to lose all teeth at once.
 - (b) Medical Constraints: When extractions must be staged over time.
3. **Time/Financial Constraints:** When a definitive prosthesis must be delayed.
4. **Immediate Aesthetics**

Immediate denture: It is a prosthesis used to replace one or more teeth and is inserted on the day of the extraction of teeth. It is a partial denture constructed before the extraction of unwanted teeth and is inserted immediately after this extraction.

Temporary improvement of esthetics after a sudden loss of teeth, before sufficient healing has occurred to fabricate a permanent prosthesis (accidents, after extractions), usually six weeks is enough for tissue healing.

- 5. Temporary space maintenance** for missing teeth. In children, the space maintainer keeps the space for the successive permanent teeth to erupt. It prevents extrusion and migration of adjacent/opposing dentition in adults.



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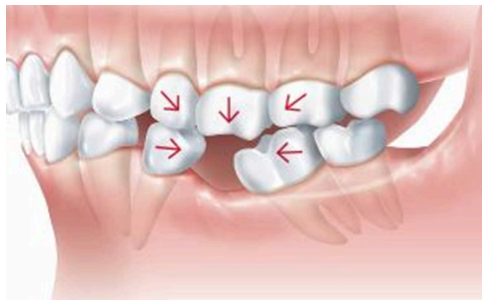


Figure 1: Consequences of tooth extraction.

- 6. Treatment Denture:** a removable partial denture used to improve a condition before a definitive denture can be made. The types of treatment may include:

- a.** Tissue Conditioning - papillary hyperplasia (massage, brushing, with or without surgery) - acute inflammation (increase tissue adaptation and redistribute the stress) - may use the existing denture, or a new treatment denture may be made



Figure 2: papillary hyperplasia from ill-fitting RPD ([Cruz-Olivo et al, 2020](#))

- b. Implant healing - a treatment RPD may be necessary for esthetics and/or function - an existing RPD or a new denture may be made - a soft liner can be placed so that the fixtures can heal unloaded.
- c. Alteration of Vertical Dimension / Occlusal Scheme - important use to determine how the patient will react to the changes (TMJ) - may be required to use an occlusal splint.



Figure 3: alteration of vertical dimension. ([Fabbri et al, 2018](#))

- d. Surgical Splint - after the removal of palatal tori, etc

Components of Interim Partial Denture

1. **Artificial teeth**
2. **Denture base:** The base acts as a saddle and as a connector and has a reciprocal function for the retentive arms. Full palatal coverage increases strength and stability. Generally, the denture is extended to at least the first molar for stability. The denture base is usually made of acrylic resin or polyamide flexible dentures.



3. **Direct retainers**
 - A. Flexible clasps in flexible dentures.
 - B. Wrought stainless steel wire clasps are used with acrylic partial dentures; they are attached to the acrylic denture base by embedding their non-retentive portion into the denture base.

The most common clasp design used for acrylic partial dentures is a **simple circular clasp**. It is used for the teeth adjacent to the edentulous ridges. It starts lingually and passes over the ridge along the proximal surface of the abutment tooth to engage buccal undercuts. It should pass 3-4 mm away from the proximal surface of the clasped tooth to allow grinding.



Figure 4: a,b, flexible clasps. c, stainless steel clasp.

Drawbacks of removable partial denture

1. Acrylic dentures are often thicker and more noticeable than metal dentures, which can affect aesthetics.
2. Acrylic is less durable than metal, making it more susceptible to cracks, chips, and fractures.
3. The bulk of acrylic dentures can increase the risk of tissue trauma and irritation.
4. Over time, the pressure from acrylic dentures can contribute to alveolar ridge resorption, leading to changes in the shape and size of the jawbone



Figure 5: a, b) Caries and gum stripping caused by a poorly supported denture and a lingual plate with a poor fit of collets.